

APRIOPRIATE ALGORITHM

OVERVIEW

- Algorithm used for finding frequent itemsets and generating association rules
- Algorithm uses an iterative approach known as level-wise search, where k-itemsets (itemsets of size k) are used to explore (k+1)-itemsets.

EXAMPLE : GROCERY STORE

Step 1 : Initialization (Generate 1-itemset)

- List all items and their support count

Item	Support Count
Bread	4
Milk	4
Diaper	4
Beer	3
Eggs	1
Coke	2

Step 2 : Prune (Prune 1-itemsets)

- Remove items that do not meet the minimum thresh

Item	Support Count
Bread	4
Milk	4
Diaper	4
Beer	3

Step 3 : Join Step (Generate 2-itemsets)

- Form pairs of the remaining items and calculate their support

Step 4 : Prune Step (Prune 2-itemsets)

- Remove itemsets that do not meet the minimum threshold (support < 3)

Step 5 : Repeat (Generate 3-itemsets)

- Form sets of 3 items from the frequent 2-itemsets and calculate their support.

Step 6 : Repeat (Generate 3-itemsets)

- Remove itemsets that do not meet the minimum support threshold (support < 3).
- No 3-itemsets meet the minimum support threshold. Hence, the process stops here.

Step 6 : Repeat (Generate 3-itemsets)

- From the frequent itemsets, generate association rules with minimum confidence of 80%

ADVANTAGES

Uses large itemset property

Easily parallelized

Easy to implement

DISADVANTAGES

Assumes transaction database is memory resident.

Requires many database scans.

Uses a uniform minimum support threshold

Applications of Association Rules

Retail: Product recommendations.

Healthcare: Correlating symptoms & treatments.

Web Usage: Suggesting related pages.

E.g. : Placing bread near butter, to increase sales.

Challenges in Association Rule Mining

Scalability

Exponential growth of candidates.

Noise

Irrelevant frequent items (e.g. seasonal products).

Scalability

Choosing the right support/confidence thresholds.



Name: LING YU QIAN
Matric No. : A23CS0301
Email: lingqian@graduate.utm.my



Name: HARINI A/P SANGARAN
Matric No. : A23CS0081
Email: harini04@graduate.utm.my



Name: NURUL ADRIANA BINTI KAMAL JEFRI
Matric No. : A23CS0258
Email: adriana04@graduate.utm.my